Credit Risk Analysis

Lending Club Loan Portfolio | 2007 - 2018

An end-to-end analysis of 1.35 million completed Lending Club loans to identify what drives borrower default, quantify default rates across loan attributes, and segment the portfolio into Low / Medium / High risk tiers for lending and pricing decisions.

1.35M	**19.96%**	**6.0% -> 49.9%**	**34.99%**
Loans analyzed	Overall default rate	Grade A -> G	High-risk segment

Tools: Python (pandas, NumPy, matplotlib, seaborn) | SQL (MySQL) | Power BI | Statistics

Data source: Kaggle - Lending Club accepted loans (2007-2018)

Report date: 22 June 2026

1. Executive Summary

Business question: Which borrowers are likely to default, and how do we group them into risk tiers? A lender uses this to decide who to approve, how to price loans, and which accounts to monitor more closely.

This project analyses 1,345,350 completed Lending Club loans (those that either fully paid or charged off / defaulted) issued between 2007 and 2018. Across the portfolio, 268,599 loans defaulted - an overall default rate of 19.96%. Default risk rises sharply and predictably with loan grade, debt-to-income (DTI), loan term, and loan purpose, and falls with income. Combining these signals, a simple rule-based risk score splits the book into three tiers whose default rates differ by almost 3x.

Headline findings

- Overall default rate of 19.96% across 1.35M completed loans.
- Grade is the single strongest signal: Grade A defaults at 6.04% vs Grade G at 49.93% - roughly 8.3x higher.
- 60-month loans default at 32.45% vs 16.00% for 36-month loans (2x).
- Riskiest purpose is small business (29.71%); safest is wedding (12.16%) and car (14.70%).
- Borrowers with DTI above 40 default at 30.55% vs 14.89% for DTI 0-10.
- The High-risk segment defaults at 34.99% vs 12.42% for Low-risk, while carrying USD 2.35B of exposure.

2. Dataset & Methodology

2.1 Data source

Lending Club accepted-loans dataset (Kaggle: wordsforthewise/lending-club). The raw file contains 2,260,701 loan records and 150+ columns. 22 columns relevant to credit risk were loaded to keep memory manageable.

2.2 Defining the target: default

Only loans with a final outcome were kept. Loans marked Fully Paid were treated as good (0); loans marked Charged Off or Default were treated as bad (1). In-progress statuses (Current, Late, In Grace Period) were removed because their final outcome is unknown. This left 1,345,350 loans.

2.3 Data cleaning

- term: ' 36 months' / ' 60 months' converted to numeric 36 / 60.
- emp_length: '10+ years' -> 10, '< 1 year' -> 0, etc.
- issue_d: parsed to a date; issue_year extracted for trend analysis.
- fico: single score taken as the midpoint of fico_range_low/high.
- int_rate, revol_util, dti, annual_inc coerced to numeric.
- Continuous fields binned into DTI bands and income bands.

2.4 Rule-based risk score

A transparent 0-6 point score adds one or two points for each risk signal, then maps the total to a segment - Low (0-1), Medium (2-3), High (4+):

- Grade C/D: +1   Grade E/F/G: +2
- DTI above 25: +1
- FICO below 680: +1
- Any delinquency in last 2 years: +1
- Revolving utilisation above 60%: +1

2.5 Workflow

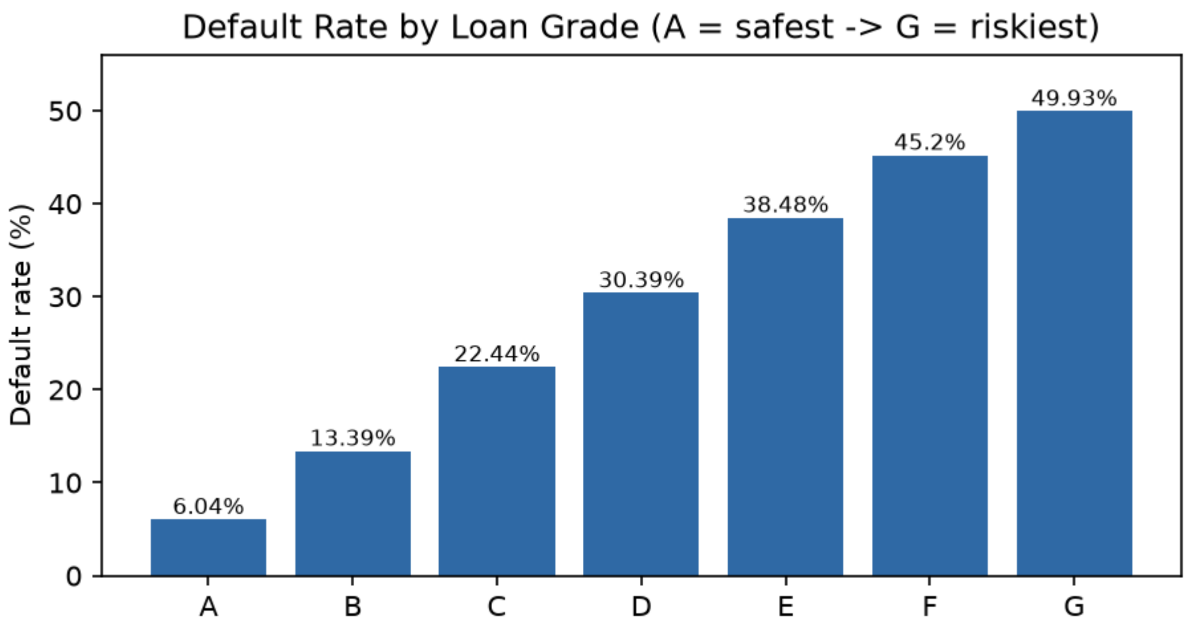
Python (pandas) for cleaning, target creation, analysis and scoring -> cleaned data exported to CSV -> loaded into MySQL where 9 analysis queries reproduce the KPIs -> the cleaned dataset visualised in an interactive Power BI dashboard.

3. Key Findings

3.1 Default rate by loan grade

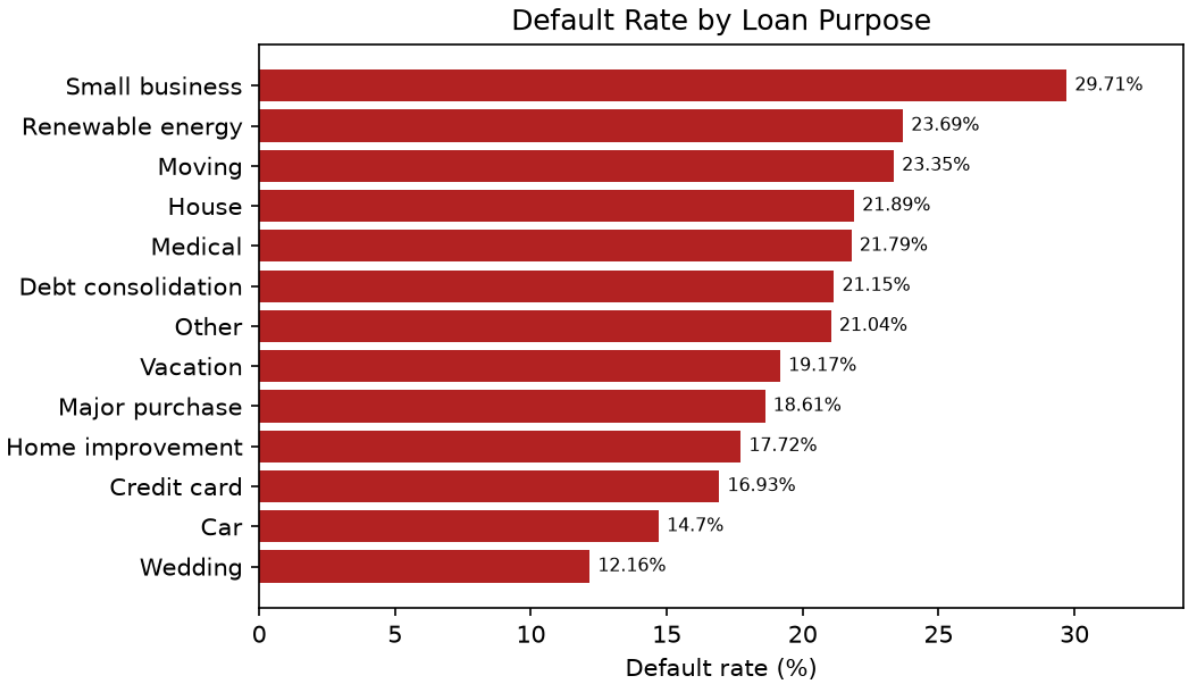
Default rate climbs monotonically from Grade A to G, and the average interest rate rises with it - confirming risk is priced into the rate.

Grade	Loans	Default rate	Avg int. rate
A	235,095	6.04%	7.11%
B	392,748	13.39%	10.68%
C	381,694	22.44%	14.02%
D	200,966	30.39%	17.72%
E	93,656	38.48%	21.14%
F	32,059	45.2%	24.93%
G	9,132	49.93%	27.73%



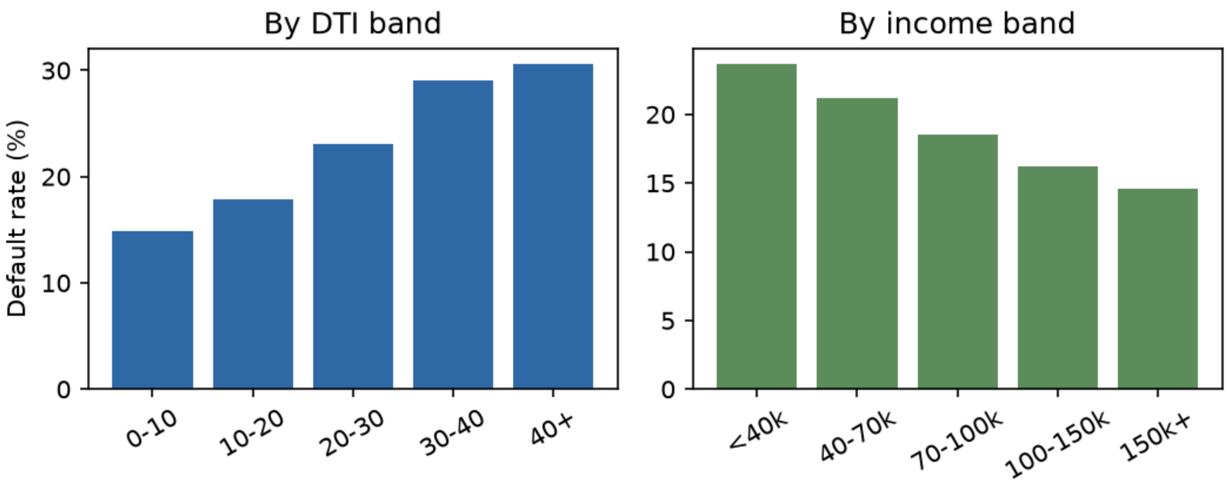
3.2 Default rate by loan purpose

Small-business loans are clearly the riskiest use of funds; weddings, cars and credit-card refinancing are the safest.



3.3 DTI and income

Default risk rises steadily with debt-to-income and falls steadily with annual income - two of the most intuitive affordability signals.



3.4 DTI and income - detail

DTI band	Loans	Default rate
0-10	246,186	14.89%
10-20	562,128	17.85%
20-30	408,353	23.05%
30-40	121,541	29.1%
40+	6,766	30.55%

Income band	Loans	Default rate
<40k	245,589	23.63%
40-70k	526,360	21.16%
70-100k	322,993	18.52%
100-150k	178,922	16.18%
150k+	71,125	14.58%

3.5 Loan term and home ownership

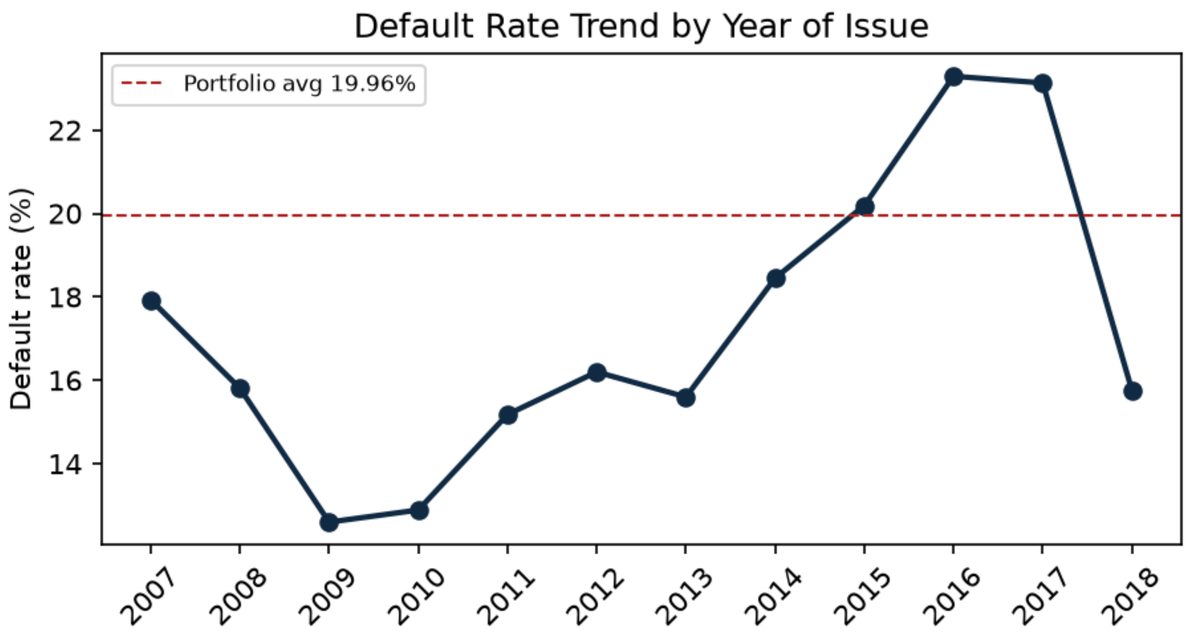
Longer 60-month loans default at twice the rate of 36-month loans. Renters default more often than owners and mortgage-holders.

Term	Loans	Default rate
36 months	1,020,768	16.0%
60 months	324,582	32.45%

Home ownership	Loans	Default rate
RENT	534,436	23.22%
OWN	144,840	20.62%
MORTGAGE	665,596	17.21%

3.6 Default rate trend by year

Default rates rose through 2015-2016 as the book grew rapidly. Recent vintages (2017-2018) contain loans that may not have fully matured, so their rates should be read with caution.



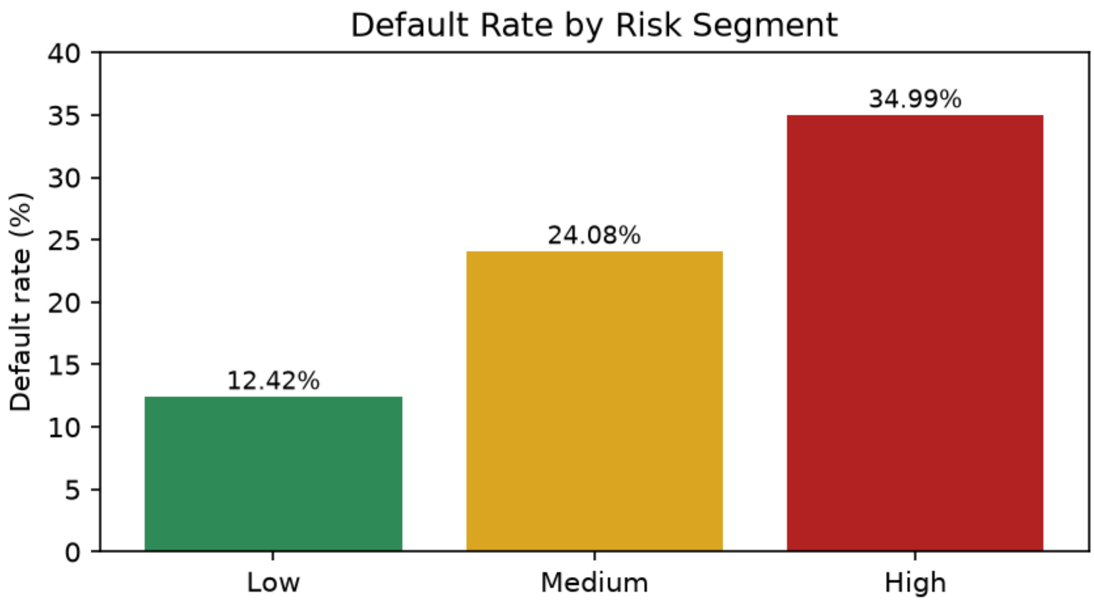
3.7 Highest-risk states (min. 1,000 loans)

#	State	Default rate
1	MS	26.08%
2	NE	25.18%
3	AR	24.09%
4	AL	23.63%
5	OK	23.48%
6	LA	23.18%
7	NY	22.05%
8	NV	21.92%
9	FL	21.48%
10	TN	21.41%

4. Risk Segmentation (key deliverable)

The rule-based score groups every loan into Low, Medium or High risk. The three tiers separate cleanly on default rate, interest rate and exposure - giving the lender an actionable way to triage the portfolio.

Segment	Loans	Default rate	Avg int. rate	Exposure (USD)
Low	613,598	12.42%	10.25%	8.58B
Medium	583,349	24.08%	14.98%	8.46B
High	148,403	34.99%	18.77%	2.35B



The High-risk tier holds 148K loans (11% of the book) yet defaults at 34.99% - nearly 3x the Low-risk tier - and represents USD 2.35B of exposure that warrants the closest monitoring.

5. Business Recommendations

Approve / decline

- Tighten approvals for Grade E-G applicants, especially where DTI > 25 or FICO < 680 - these stack into the High-risk segment.
- Scrutinise small-business and other high-purpose-risk applications with additional documentation.

Price for risk

- Continue pricing interest rate to grade; ensure 60-month loans carry a premium reflecting their ~2x default rate.

Monitor

- Review the High-risk segment monthly; it concentrates 35% default risk and USD 2.35B exposure in 11% of loans.
- Track default-rate trend by vintage to catch deterioration early.

Limitations

- Analysis uses completed loans only; very recent vintages are not fully matured and may understate eventual defaults.
- The risk score is a transparent rule-based heuristic, not a fitted statistical model - a logistic regression / gradient-boosted model is the natural next step.

6. Tools & Skills Demonstrated

Area	Tools / techniques
Data cleaning	Python, pandas, NumPy
Analysis	Grouped default rates, binning, correlation
SQL	MySQL - 9 analysis queries reproducing the KPIs
Visualisation	matplotlib, seaborn, Power BI dashboard
Statistics	Risk scoring, segmentation, trend analysis
Reporting	This PDF report + interactive dashboard

Generated from credit_risk_analysis.ipynb and the MySQL query results. Dataset: Kaggle - Lending Club (2007-2018).